2025-2026 Spring
MAT124 Final
08/06/2026

1. Use the method of Lagrange multipliers to find the absolute maximum and absolute minimum values of the function $f(x, y, z) = x^2 + 2y - z^2$ subject to the constraint $2x^2 + y^2 + z^2 = 4$.

2. Consider the vector field

$$\mathbf{F}(x, y, z) = (2xe^y + z\cos(xz))\mathbf{i} + (x^2e^y - \sin(y))\mathbf{j} + (x\cos(xz) + 2z)\mathbf{k}.$$

a. Show that $\mathbf{F}$ is a conservative vector field by verifying the cross-partial derivative conditions.

b. Find a potential function $\phi(x, y, z)$ such that $\mathbf{F} = \nabla\phi$.

c. Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the path given by $\mathbf{r}(t) = \left\langle t, t(1-t), \frac{\pi}{2}t^3 \right\rangle$ for $0 \leq t \leq 1$.

3. Let C be the upper half of the circle $x^2 + y^2 = 9$, oriented from $(3, 0)$ to $(-3, 0)$. Evaluate the line integral

$$\int_C \left(y^2 + \sin(x^3)\right) dx + \left(3xy + e^{y^2}\right) dy.$$

Hint: The curve C is not closed. Consider adding a line segment to form a closed loop and use Green's Theorem.

4. Answer the following questions. For multiple choice questions, mark the correct option. For others, write your answer in the box provided. No partial credits will be given.

4.1. Consider the iterated double integral $\int_{-8}^{8} \int_{x^{2/3}}^{4} f(x, y)\, dy\, dx$. Reverse the order of integration and write the new limits in the boxes below:

$$\int_{\boxed{}}^{\boxed{}} \int_{\boxed{}}^{\boxed{}} f(x, y)\, dx\, dy$$

4.2. Let E be the solid region bounded above by the paraboloid $z = x^2 + y^2$, below by the xy-plane ($z = 0$), and enclosed by the elliptic cylinder $x^2 + 4y^2 = 4$. What is the volume of the region E?
(Hint: Use a modified coordinate system such as $x = 2r\cos\theta$ and $y = r\sin\theta$ to simplify the region).

(A) $\dfrac{3\pi}{2}$ (B) 2π (C) $\dfrac{5\pi}{2}$ (D) 3π (E) 4π

4.3. Consider the vector field $\mathbf{F}(x, y) = \left\langle y^2, -\frac{2}{x-2} \right\rangle$ representing the velocity field of a steady fluid flow. Imagine placing a tiny paddle wheel into this fluid. The paddle wheel will rotate counter-clockwise if the scalar curl of the field is positive, and clockwise if it is negative. In which of the following regions above the x-axis ($y > 0$) will the paddle wheel definitely rotate in a counter-clockwise direction?

(A) $0 < y < \dfrac{1}{(x-2)^2}$

(B) $y^2 > \dfrac{2}{x-2}$

(C) $y^2 < \dfrac{2}{x-2}$

(D) $y > \dfrac{1}{(x-2)^2}$

(E) Everywhere above the x-axis.

4.4. Let D be the region in the xy-plane bounded by the lines $y = x$, $y = x - 2$, $y = -x$, and $y = -x + 3$. Using an appropriate change of variables, evaluate the double integral:

$$\iint_D (x + y)e^{x^2 - y^2}\, dA$$

(A) $\frac{1}{2}(e^6 - 1)$ (B) $\frac{1}{4}(e^6 - 7)$ (C) $e^6 - 7$ (D) $\frac{3}{2}(e^4 - 1)$ (E) $2(e^6 - 3)$

4.5. Which of the following curves represents a field line (integral curve) for the vector field $\mathbf{F}(x, y) = y\mathbf{i} + x\mathbf{j}$? (where C is a constant)

(A) $x^2 + y^2 = C$ (B) $y = Cx$ (C) $y = Ce^x$ (D) $x^2 - y^2 = C$ (E) $xy = C$

4.6. Calculate the outward flux of the vector field $\mathbf{F}(x, y, z) = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$ across the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

(A) $\dfrac{4\pi}{5}$ (B) $\dfrac{8\pi}{5}$ (C) $\dfrac{12\pi}{5}$ (D) 4π (E) 12π

4.7. Consider the following two vector fields in $\mathbb{R}^3$:

$$\mathbf{F}_1 = \langle -y, x, 0 \rangle \quad \text{and} \quad \mathbf{F}_2 = \langle x, y, z \rangle$$

How can these fields be classified?

(A) Both are solenoidal.

(B) Both are irrotational.

(C) $\mathbf{F}_1$ is solenoidal, $\mathbf{F}_2$ is irrotational.

(D) $\mathbf{F}_1$ is irrotational, $\mathbf{F}_2$ is solenoidal.

(E) Neither field is solenoidal nor irrotational.

4.8. The vector field $\mathbf{F}(x, y, z) = 2y\mathbf{i} + z\mathbf{j} + 0\mathbf{k}$ is solenoidal. Find a vector potential $\mathbf{A}$ such that $\nabla \times \mathbf{A} = \mathbf{F}$. Assume the x-component of the vector potential is zero. Write the remaining components.

1. We need to find the extrema of $f(x,y,z) = x^2 + 2y - z^2$ subject to $g(x,y,z) = 2x^2 + y^2 + z^2 = 4$. The Lagrange multiplier equations are $\nabla f = \lambda \nabla g$ and $g = 4$.

$$\begin{aligned} 2x &= 4\lambda x \implies 2x(1-2\lambda) = 0 && (1)\\ 2 &= 2\lambda y \implies \lambda y = 1 && (2)\\ -2z &= 2\lambda z \implies 2z(1+\lambda) = 0 && (3)\\ 2x^2 + y^2 + z^2 &= 4 && (4) \end{aligned}$$

From equation (3), we have two possible cases: $z = 0$ or $\lambda = -1$.

Case 1: $z = 0$
From equation (1), either $x = 0$ or $\lambda = \frac{1}{2}$.

- **Subcase 1a:** $x = 0$
  Substituting $x = 0$ and $z = 0$ into (4): $0 + y^2 + 0 = 4 \implies y = \pm 2$.
  This gives the points $(0, 2, 0)$ and $(0, -2, 0)$.
  Evaluating the function:
  $$f(0,2,0) = 4 \quad \text{and} \quad f(0,-2,0) = -4.$$

- **Subcase 1b:** $\lambda = \frac{1}{2}$
  From (2), $y = \frac{1}{\lambda} = 2$.
  Substituting $y = 2$ and $z = 0$ into (4): $2x^2 + 4 + 0 = 4 \implies 2x^2 = 0 \implies x = 0$.
  This yields the point $(0, 2, 0)$ again.

Case 2: $\lambda = -1$
From (2), $y = \frac{1}{-1} = -1$.
From (1), $2x(1 - 2(-1)) = 6x = 0 \implies x = 0$.
Substituting $x = 0$ and $y = -1$ into (4): $0 + (-1)^2 + z^2 = 4 \implies z^2 = 3 \implies z = \pm\sqrt{3}$.
This gives the points $\left(0, -1, \sqrt{3}\right)$ and $\left(0, -1, -\sqrt{3}\right)$.
Evaluating the function:

$$f\left(0, -1, \pm\sqrt{3}\right) = 0^2 + 2(-1) - \left(\pm\sqrt{3}\right)^2 = -2 - 3 = -5.$$

Compare the values evaluated at all critical points.

$$\boxed{\text{Absolute maximum: } 4 \text{ at } (0,2,0), \text{ absolute minimum: } -5 \text{ at } \left(0,-1,\pm\sqrt{3}\right).}$$

2. **a.** Let $\mathbf{F} = \langle P, Q, R\rangle$. We check the curl components:

$$\begin{aligned}
&P_y = 2xe^y, && Q_x = 2xe^y && \implies P_y = Q_x\\
&P_z = \cos(xz) - xz\sin(xz), && R_x = \cos(xz) - xz\sin(xz) && \implies P_z = R_x\\
&Q_z = 0, && R_y = 0 && \implies Q_z = R_y
\end{aligned}$$

Since the domain is all of $\mathbb{R}^3$ (which is simply connected) and all cross-partials are equal, $\mathbf{F}$ is conservative.

b. We need ϕ such that $\phi_x = P$, $\phi_y = Q$, $\phi_z = R$.

$$\phi(x, y, z) = \int (2xe^y + z\cos(xz))\,dx = x^2e^y + \sin(xz) + g(y, z).$$

Differentiating with respect to y,

$$\phi_y = x^2e^y + g_y(y, z) = Q = x^2e^y - \sin(y),$$

$$\implies g_y(y, z) = -\sin(y) \implies g(y, z) = \cos(y) + h(z).$$

Thus $\phi(x, y, z) = x^2e^y + \sin(xz) + \cos(y) + h(z)$. Differentiating with respect to z,

$$\phi_z = x\cos(xz) + h'(z) = R = x\cos(xz) + 2z \implies h'(z) = 2z \implies h(z) = z^2.$$

Thus the potential function is

$$\boxed{\phi(x, y, z) = x^2e^y + \sin(xz) + \cos(y) + z^2}.$$

c. By the Fundamental Theorem of Line Integrals, $\int_C \mathbf{F}\cdot d\mathbf{r} = \phi(\mathbf{r}(1)) - \phi(\mathbf{r}(0))$.

- Start point: $\mathbf{r}(0) = \langle 0, 0, 0\rangle \implies \phi(0, 0, 0) = 0 + 0 + \cos(0) + 0 = 1$.
- End point: $\mathbf{r}(1) = \left\langle 1, 0, \frac{\pi}{2}\right\rangle \implies \phi\left(1, 0, \frac{\pi}{2}\right) = 1^2e^0 + \sin\left(\frac{\pi}{2}\right) + \cos(0) + \left(\frac{\pi}{2}\right)^2 = 1 + 1 + 1 + \frac{\pi^2}{4} = 3 + \frac{\pi^2}{4}$.

The integral is

$$\boxed{\left(3 + \frac{\pi^2}{4}\right) - 1 = 2 + \frac{\pi^2}{4}}.$$

3. The curve C is the upper half of the circle of radius 3, oriented from $(3,0)$ to $(-3,0)$. This is not a closed curve. Let C_1 be the line segment along the x-axis from $(-3,0)$ to $(3,0)$.

The combined curve $C \cup C_1$ forms a closed loop K traversing the boundary of the upper half-disk D in the positive (counter-clockwise) direction. By Green's Theorem,

$$\oint_K P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Here, $P = y^2 + \sin(x^3)$ and $Q = 3xy + e^{y^2}$.

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 3y - 2y = y$$

We evaluate the double integral over the upper half-disk D using polar coordinates $(0 \le r \le 3,\ 0 \le \theta \le \pi)$.

$$\iint_D y\,dA = \int_0^\pi \int_0^3 (r\sin\theta) r\,dr\,d\theta = \left(\int_0^\pi \sin\theta\,d\theta\right)\left(\int_0^3 r^2\,dr\right)$$

$$= [-\cos\theta]_0^\pi \cdot \left[\frac{r^3}{3}\right]_0^3 = (1-(-1)) \cdot 9 = 18.$$

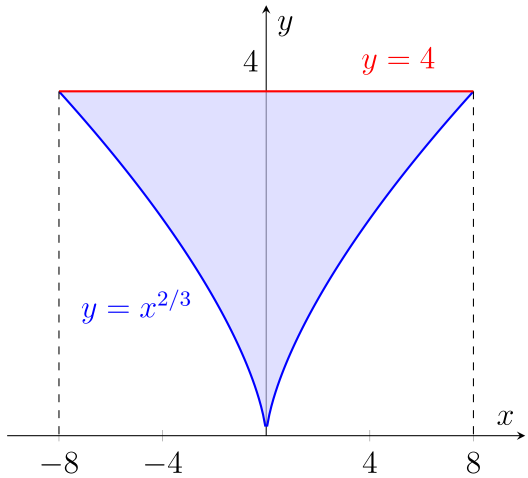
So, $\int_C \mathbf{F}\cdot d\mathbf{r} + \int_{C_1} \mathbf{F}\cdot d\mathbf{r} = 18$. Now we evaluate the line integral over C_1. On C_1, $y = 0 \implies dy = 0$, and x goes from -3 to 3:

$$\int_{C_1} P\,dx + Q\,dy = \int_{-3}^3 \sin(x^3)\,dx.$$

Since $f(x) = \sin(x^3)$ is an odd function $(f(-x) = -f(x))$ and the interval $[-3,3]$ is symmetric around the origin, this integral is strictly 0. Therefore,

$$\int_C \mathbf{F}\cdot d\mathbf{r} + 0 = 18 \implies \int_C \mathbf{F}\cdot d\mathbf{r} = \boxed{18}.$$

4. 4.1. The region of integration is defined by $-8 \leq x \leq 8$ and $x^{2/3} \leq y \leq 4$. The lower bound curve is $y = x^{2/3}$. Solving for x gives $y^3 = x^2$, which means $x = \pm y^{3/2}$. Since x ranges from -8 to 8, the new horizontal bounds for a fixed y are from the left branch $x = -y^{3/2}$ to the right branch $x = y^{3/2}$. The total range for y is from 0 to 4.



$$\boxed{\int_0^4 \int_{-y^{3/2}}^{y^{3/2}} f(x, y)\, dx\, dy}$$

4.2. Using the suggested generalized polar coordinates $x = 2r\cos\theta$ and $y = r\sin\theta$. The Jacobian of this transformation is $J = 2r$. The boundary of the region D is the elliptic cylinder $x^2+4y^2 = 4$ which simplifies to $4r^2\cos^2\theta+4r^2\sin^2\theta = 4 \Rightarrow r = 1$ Thus, $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$.

$$\begin{aligned}
V &= \iint_D (x^2 + y^2)\, dA = \int_0^{2\pi} \int_0^1 \left((2r\cos\theta)^2 + (r\sin\theta)^2\right)(2r)\, dr\, d\theta \\
&= \int_0^{2\pi} \int_0^1 2r^3(4\cos^2\theta + \sin^2\theta)\, dr\, d\theta \\
&= \left(\int_0^1 2r^3\, dr\right)\left(\int_0^{2\pi} (4\cos^2\theta + \sin^2\theta)\, d\theta\right) \\
&= \left[\frac{r^4}{2}\right]_0^1 \cdot (4\pi + \pi) = \frac{1}{2} \cdot 5\pi = \frac{5\pi}{2}
\end{aligned}$$

$\boxed{\text{Answer: (C)}}$

4.3. For the paddle wheel to rotate counter-clockwise, the scalar curl (z-component of the curl) must be positive.

$$\text{curl}\,\mathbf{F} \cdot \mathbf{k} = \frac{\partial}{\partial x}\left(-\frac{2}{x-2}\right) - \frac{\partial}{\partial y}(y^2) = \frac{2}{(x-2)^2} - 2y$$

Setting the curl greater than zero:

$$\frac{2}{(x-2)^2} - 2y > 0 \Rightarrow 2y < \frac{2}{(x-2)^2} \Rightarrow y < \frac{1}{(x-2)^2}$$

Since the question asks for the region above the x-axis, we also have $y > 0$.

$\boxed{\text{Answer: (A)}}$

4.4. Let $u = x - y$ and $v = x + y$. The bounding lines become $u = 0$, $u = 2$, $v = 0$, and $v = 3$.

Solving for x and y, we get $x = \frac{1}{2}(u + v)$ and $y = \frac{1}{2}(v - u)$.

The absolute value of the Jacobian determinant is $\frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{2}$.

The integrand becomes $(x + y)e^{x^2-y^2} = ve^{uv}$.

$$\int_0^3 \int_0^2 ve^{uv}\left(\frac{1}{2}\right) du\ dv = \frac{1}{2}\int_0^3 [e^{uv}]_{u=0}^{u=2} dv = \frac{1}{2}\int_0^3 (e^{2v} - 1)dv$$

$$= \frac{1}{2}\left[\frac{1}{2}e^{2v} - v\right]_0^3 = \frac{1}{2}\left(\frac{1}{2}e^6 - 3 - \left(\frac{1}{2} - 0\right)\right)$$

$$= \frac{1}{2}\left(\frac{1}{2}e^6 - \frac{7}{2}\right) = \frac{1}{4}(e^6 - 7)$$

$\boxed{\text{Answer: (B)}}$

4.5. The field lines satisfy the differential equation $\frac{dy}{dx} = \frac{Q(x,y)}{P(x,y)}$.

$$\frac{dy}{dx} = \frac{x}{y} \Rightarrow y\ dy = x\ dx$$

Integrating both sides yields $\frac{1}{2}y^2 = \frac{1}{2}x^2 + K \Rightarrow x^2 - y^2 = C$

$\boxed{\text{Answer: (D)}}$

4.6. By the Divergence Theorem, the flux is $\iiint_E \operatorname{div} \mathbf{F}\ dV$.

$$\operatorname{div} \mathbf{F} = 3x^2 + 3y^2 + 3z^2 = 3\rho^2.$$

Using spherical coordinates for the unit sphere $(0 \le \rho \le 1, 0 \le \phi \le \pi, 0 \le \theta \le 2\pi)$.

$$\text{Flux} = \int_0^{2\pi}\int_0^{\pi}\int_0^1 (3\rho^2)\rho^2 \sin\phi\ d\rho\ d\phi\ d\theta$$

$$= 3\left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi} \sin\phi\ d\phi\right)\left(\int_0^1 \rho^4 d\rho\right) = 3 \cdot (2\pi) \cdot (2) \cdot \left(\frac{1}{5}\right) = \frac{12\pi}{5}$$

$\boxed{\text{Answer: (C)}}$

4.7. For $F_1 = \langle -y, x, 0\rangle$: $\operatorname{div}\mathbf{F}_1 = 0+0+0 = 0$ (Solenoidal). $\operatorname{curl}\mathbf{F}_1 = \langle 0,0,2\rangle \neq 0$
For $\mathbf{F}_2 = \langle x, y, z\rangle$: $\operatorname{div}\mathbf{F}_2 = 1+1+1 = 3 \neq 0$. $\operatorname{curl}\mathbf{F}_2 = \langle 0,0,0\rangle$ (Irrotational).
$\boxed{\text{Answer: (C)}}$

4.8. We are looking for $\mathbf{A} = \langle 0, \mathbf{A}_2, \mathbf{A}_3\rangle$ such that $\nabla \times \mathbf{A} = \langle 2y, z, 0\rangle$.

$$\nabla \times \mathbf{A} = \left\langle \frac{\partial \mathbf{A}_3}{\partial y} - \frac{\partial \mathbf{A}_2}{\partial z}, -\frac{\partial \mathbf{A}_3}{\partial x}, \frac{\partial \mathbf{A}_2}{\partial x} \right\rangle$$

Equating components:

- $-\dfrac{\partial \mathbf{A}_3}{\partial x} = z \Rightarrow \mathbf{A}_3 = -xz + g(y,z)$
- $\dfrac{\partial \mathbf{A}_2}{\partial x} = 0 \Rightarrow \mathbf{A}_2 = h(y,z)$
- $\dfrac{\partial \mathbf{A}_3}{\partial y} - \dfrac{\partial \mathbf{A}_2}{\partial z} = 2y$

Substitute $\mathbf{A}_3$ and $\mathbf{A}_2$ into the third equation: $\frac{\partial g}{\partial y} - \frac{\partial h}{\partial z} = 2y$.
There are infinitely many valid choices. A simple choice is to let $h(y,z) = 0$ (so $\mathbf{A}_2 = 0$) Then $\frac{\partial g}{\partial y} = 2y \Rightarrow g(y,z) = y^2$. This yields $\mathbf{A}_3 = y^2 - xz$.
Another valid choice is letting $g(y,z) = 0$ (so $\mathbf{A}_3 = -xz$). $-\frac{\partial h}{\partial z} = 2y \Rightarrow h(y,z) = -2yz$. This yields $\mathbf{A}_2 = -2yz$.
Answer: (Using the first choice)

$$\boxed{\mathbf{A}(x,y,z) = \langle 0, 0, y^2 - xz\rangle}$$

(Note: $\langle 0, -2yz, -xz\rangle$ is also a correct valid pair). Other valid answers are also welcome.